UNITED STATES SPACE FORCE

Space Launch Delta 45

Spaceport of the Future Infrastructure (SPotFi)

REQUEST FOR INFORMATION (RFI) / MARKET RESEARCH NOTICE

Sonic Boom and Launch Noise Analysis

Material Testing Methodology and Predictive Engineering Model

Cape Canaveral Space Force Station (CCSFS), Florida

Date of Issuance	10 November 2026
Response Due	3 December 2026 | 2:00 PM EST
NAICS Code	1. 541330 — Engineering Services 2. 541715 — Research and Development in the Physical, Engineering, and Life Sciences (except Nanotechnology and Biotechnology)
PSC Code	1a. R425 — Support—Professional: Engineering/Technical 2a. AR12 — Space R&D Services; Space Flight, Research and Supporting Activities – Applied Research
Small Business Size Standard	1b. $25.5M 2b. 1000 Employees
Place of Performance	Primarily contractor facilities / laboratories, with potential on-site work at CCSFS, Florida. Model development and analysis may be performed at contractor location.
Anticipated Period of Performance	36 months base period from date of award, with one 12-month option period.
Anticipated Contract Type	Firm Fixed Price

1. Nature of This Announcement

This is a Request for Information (RFI) for market intelligence and for informational purposes only. This announcement does not constitute a solicitation or a request for proposals (RFP), and no solicitation is currently available. Participation in this RFI is strictly voluntary. The Government will not be obligated to award any agreement or contract because of this RFI, nor will it reimburse respondents for any costs associated with the preparation or submission of information. Submitting a response will not affect a firm's ability to submit a proposal should a formal solicitation be issued in the future.

Responses will be used to: (a) assess industry interest and capability; (b) identify the engineering disciplines under which this work is most appropriately performed and classified; (c) refine technical requirements and acquisition strategy; and (d) inform NAICS, PSC, set-aside, and contract-type decisions.

NOTE: While Other Transaction Authority, if used, is not subject to FAR-based small business regulations, the NAICS code is provided for market research to help the Government understand the size and demographics of the interested industrial base.

Do not submit proprietary trade secrets, protected technologies, or classified information in your response. All feedback received will be handled as public market research to shape the Government's final acquisition approach. Limit responses to unclassified information. If CUI handling capabilities are relevant, describe them at an unclassified level only.

2. Program / Project Overview

Facilities and infrastructure at Cape Canaveral Space Force Station (CCSFS) are exposed to intense acoustic loads and sonic booms from an increasing cadence of launch operations. Existing historical data is primarily based on aircraft overflights and has limited applicability to the unique pressure signatures generated by modern launch vehicles (including Falcon 9, Falcon Heavy, Starship/Super Heavy, Vulcan, New Glenn, and other current or proposed vehicles).

The Government requires development of a scientifically defensible material testing methodology and a predictive engineering model to evaluate the effects of launch noise and sonic booms on both current and legacy construction materials, infrastructure, and facility types at CCSFS. The effort shall support informed decisions on facility sustainment, retrofitting, or replacement. Reference the attached Statement of Objective for a more detailed description of this requirement.

Objective. Develop, validate, and deliver a repeatable material testing method and an engineering model that predicts the structural response and damage potential to CCSFS facilities and infrastructure from a wide variety of current and future launch vehicle acoustic environments and sonic booms as a function of distance, event type, and vehicle configuration.

3. Purpose of This RFI

The Government is conducting market research to:

- Determine the engineering discipline(s) under which this work most appropriately falls, and the typical team composition required to perform it.
- Identify firms with the laboratory, analytical, and subject-matter capability to develop a scientifically defensible test method that replicates launch-noise and sonic-boom signatures and to produce a validated predictive model.
- Understand industry's view of the correct NAICS/PSC pairing, small-business implications, and realistic contract type.
- Identify technical risks, existing methods or data that could be leveraged, and any constraints (test-article scale, facility limitations, data rights, CUI handling).
- Inform the Government's acquisition strategy, including potential set-aside, teaming, and evaluation approach.

4. Information Requested

Please respond to the questions below if interested no later than **3 December 2026, 2:00 PM EST.** You may address questions in a narrative, annotated outline, bullet or table format.

Do not include classified or proprietary information.

A. Engineering Disciplines

Q1. Identify the PRIMARY engineering discipline, list supporting/secondary engineering disciplines, scientific disciplines, and provide recommended labor categories that apply for this effort.

Examples (non-exhaustive): Acoustical Engineering; Structural Engineering; Materials Science / Materials Engineering; Mechanical Engineering; Aerospace Engineering; Civil / Facilities Engineering; Experimental Mechanics / Test Engineering; Engineering Mechanics / Fatigue & Fracture; Dynamics & Vibration. If none of these fit, name the discipline you would use and explain.

B. Technical Capabilities — Test Method, Laboratory, and Modeling

Q2. Describe your company's testing capabilities to generate or replicate the pressure-time histories and frequency content of rocket launch noise and sonic booms.

Q3. Does your company have experience with the work required within the SOO. Please give us examples of similar types of work your firm has completed. State your interest and ability to perform this type of study.

Q4. Does your company have any limiting factors or potential performance risks to successfully complete this type of study? What do you consider the principal technical risks of this effort? How would your company mitigate identified limiting factors and risks?

D. Acquisition, Pricing, and Execution

Q5. Is the contemplated 36-month, phased, base period realistic? If not, what period of performance and phase structure would you recommend, and why?

Q6. What information would you need in a draft Statement of Objectives / PWS, or in a draft solicitation, to submit a quality proposal (e.g., representative load-case spectra, list of specific facilities, launch vehicle list, existing field data availability, specimen-supply plan)?

Q7. Provide any additional comments, alternative approaches, or recommendations not captured above that would improve the Government's probability of successful execution.

5. Submission Instructions

Submit responses by **3 December 2026 at 2:00 PM EST** via email to:

John P. Porter — john.porter.23@spaceforce.mil
Jaydin Perez — jaydin.perez.2@spaceforce.mil

- File format: Microsoft Word or searchable PDF.
- File naming Recommended: RFI_LaunchNoise_[CompanyName].docx (or .pdf).
- Email subject line Recommended: RFI Response – Sonic Boom and Launch Noise Analysis – [Company Name].

The Government may contact respondents for clarification. The Government may also conduct one-on-one industry discussions or a virtual industry day after reviewing responses; participation in this RFI does not guarantee an invitation. If interested, please request invitation

6. Government Points of Contact

Role	Contact
Contracting Officer	John P. Porter – john.porter.23@spaceforce.mil
Contracting Specialist	Jaydin Perez – jaydin.perez.2@spaceforce.mil

— END OF RFI —

The Government welcomes candid industry feedback regarding commercial practices, agreement terms, technical approach, risk sharing, data rights, and other considerations that could improve successful execution of this effort.